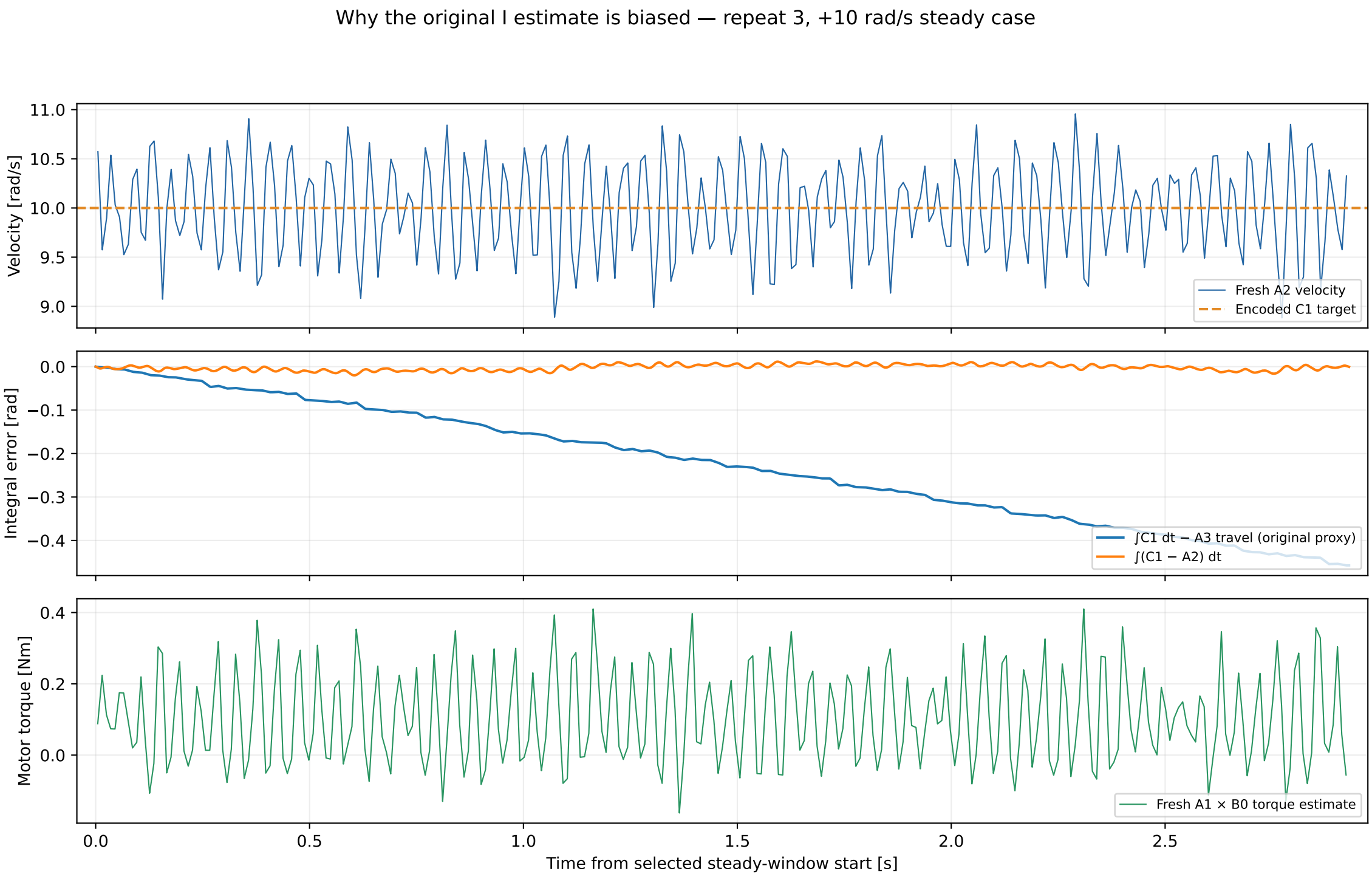
Why the original I estimate is biased — repeat 3, +10 rad/s steady case
Velocity [rad/s]
11.0
10.5
10.0
9.5
9.0
Fresh A2 velocity
Encoded C1 target
Integral error [rad]
0.0
−0.1
−0.2
−0.3
−0.4
∫C1 dt − A3 travel (original proxy)
∫(C1 − A2) dt
Motor torque [Nm]
0.4
0.2
0.0
Fresh A1 × B0 torque estimate
0.0
0.5
1.0
1.5
2.0
2.5
Time from selected steady-window start [s]
Integral origins set to zero at the analysis-window start only; hardware encoder origins are unchanged.